The Carry-Over Lottery Allocation Family: Bounds, Backtests, and a Configuration Space for NBA Draft Reform

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May 26, 2026

Abstract

The annual NBA draft is a constrained fair-allocation problem: 30 indivisible draft picks are assigned to 30 teams that take turns making selections. The top picks are allocated through a lottery for the 14 non-playoff teams. The lottery has a documented manipulation incentive ("tanking"): a team eliminated from playoff contention can improve its expected draft position (DP) by losing additional games. Banchio and Munro [1] prove an impossibility result, showing that no mechanism based on end-of-season records alone can simultaneously reward poor performance and eliminate this incentive. Carry-Over Lottery Allocation (COLA) sidesteps the impossibility by replacing single-season records with a multi-year playoff history as the basis for DP [2]. We formalize the COLA mechanism family as a configuration space parameterized by seven design dials (eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, tiebreak). Each variant proposed in Highley et al. [2] and the Substack series is a point in this space, and the NBA's 2026 "3-2-1" proposal sits at a degenerate corner. Each dial maps to a downstream league-relevant property, converting variant comparison from a prose comparison to a tuple comparison. We instantiate the framework in an empirical testbed (Section 5) that grid-searches over dial values for Pareto-optimal configurations against league-relevant objectives. As supporting analytical properties of specific dial configurations, we derive a closed-form $\sim 4\%$ upper bound on the worst-case probability gain from pool manipulation in the Classic configuration (Theorem 2), a per-series tanking bound in the Capped configuration (Lemma 3), and a 26-season historical backtest covering 1999–2025.

1 Introduction

In the 2025-26 NBA season, multiple teams were publicly accused of tanking in anticipation of a strong draft class. The accusations did not name a small minority: by some counts, roughly a third of the league's 30 teams were openly losing games at season's end to improve their lottery odds. In response, the league introduced a number of potential draft reform ideas, including the "3-2-1 lottery." [3]. The proposal includes a three-year re-evaluation provision, which means the design discussion is expected to reopen in 2029.

The annual NBA draft is a constrained fair-allocation problem. Thirty indivisible items (draft picks) are assigned to 30 agents (teams) in priority order. Picks 15–30 are assigned by reverse regular-season record. Picks 1–14 are subject to a lottery confined to the 14 non-playoff teams, with weighted odds tilted toward teams with worse records. (In the current system, the bottom three teams are weighted equally and the rest declining). Throughout the years, the NBA has adjusted odds while facing the same question: how to assign indivisible items to agents while (a) preferentially rewarding the weakest agents, (b) eliminating

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the incentive to under-perform, and (c) producing a mechanism transparent enough that fans, owners, and players can reason about it.

Banchio and Munro [1] give a formal treatment of the central tension and prove an impossibility. They define a *No-Tanking Condition* (NTC): no team can improve its expected outcome by losing. Any mechanism mapping end-of-season records to draft probabilities, while giving worse-record teams strictly better odds, must violate NTC. Only a uniform lottery (equal odds for all non-playoff teams) satisfies NTC, abandoning differential reward. The 2026 "3-2-1 lottery" proposal takes precisely this route, with adjustments for the play-in tournament and reduced odds for the three teams at the bottom of the standings[3].

Carry-Over Lottery Allocation (COLA) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [2]. Each non-playoff team accumulates draft priority over consecutive seasons; success in the playoffs or the draft itself diminishes a team's draft priority. The single-season manipulation channel that drives the Munro-Banchio counterexample becomes ineffective: no single-season choice materially shifts the multi-year ticket pool. COLA is a family of mechanisms rather than a single design; the original paper develops one concrete instance (*Classic* COLA), and subsequent work introduces variants that occupy different points in the design space.

Contributions:

- **(Section 3) A seven-dial configuration space for the COLA family.** We formalize the COLA mechanism family as a 7-tuple $(E, \Delta, \rho, W, C, S, T)$ specifying eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, and tiebreak (Definition 2). Each variant in the family (Simple, Simple Lottery, Classic, Countdown, Capped) is a point in this space, and the NBA's 2026 "3-2-1" proposal sits at a degenerate corner of the same space. Each dial maps to a downstream league-relevant property (incentive compatibility, manipulation gain, per-series cost, transparency), converting variant comparison from a prose comparison to a tuple comparison and surfacing the design degrees of freedom the variants implicitly select between. The framework is the league's design lever for the 2029 re-evaluation discussion.

- **(Section 5) A Basketball-GM empirical testbed for grid search over dial values (forthcoming).** We instantiate the configuration space in the open-source Basketball-GM simulator, varying each dial across its admissible range and reporting Pareto frontiers in dial space against league-relevant objective functions (first target: minimize the maximum years between conference finals appearances). The testbed converts the framework from a taxonomy into an optimization tool that a league can run against its own policy preferences. The current submission reports the framework and the supporting analytical properties below; the testbed sweep is in active engineering and will appear in this section by the paper deadline.

- **(Section 4 and Section 5) Supporting analytical properties: a manipulation bound and a 26-season historical backtest.** As analytical properties of specific dial configurations, we derive a closed-form upper bound of approximately 4% on the maximum probability gain from pool manipulation under reasonable distributions of draft priorities (Theorem 2), and a per-series tanking bound for the Capped configuration showing that no playoff series win ever costs a team more than $0.3 \cdot \text{MAX}$ tickets (45 at $\text{MAX} = 150$) (Lemma 3). The historical backtest computes COLA state for every team across 26 NBA seasons (1999–2025) using verified Basketball Reference data, with the 2013–2017 Philadelphia "Process" as the canonical case study: under Classic COLA, graduated diminishment dismantles the repeat-tanking strategy the current weighted lottery rewards.

2 Background and Related Work

2.1 Effort Manipulation: The Munro-Banchio Impossibility

Banchio and Munro [1] establish that any mechanism mapping end-of-season records to draft probabilities, while giving the worst team strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two eliminated teams with identical records and a final game between them. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon the goal of differentially rewarding the weakest teams.

The authors propose *T-IC* (Truncated Incentive Compatibility) as an alternative. T-IC tracks each team's conditional probability of finishing last and freezes allocations at the moment IC would first be violated. The mechanism is optimal in theory but has three practical issues. First, it is opaque to fans. Second, calculating when IC would first be violated requires precise team-level valuations of playoff-versus-draft trade-offs that the league does not have access to. Third, the truncation timing assumes teams have an incentive to win at the start of the season, but some teams have an incentive to tank from the first game.

2.2 Preference Manipulation and Axiomatic Limits

A second impossibility addresses preference misreporting (a team claiming to want a player it does not actually want, to manipulate the allocation order): Coreno and Balbuzanov [4] prove that no draft rule simultaneously satisfies Respect for Priority [5], Envy-Free up to One Object [6, 7], and Strategy-Proofness. The result is single-period and applies to one-shot allocations of indivisible items. Whether the axioms lift to the multi-year priority structure of COLA is open work outside the scope of this paper; we note the parallel impossibility here as a complement to the Munro-Banchio effort-manipulation channel that motivates the bound in Section 4.

2.3 Prior Reform Proposals

Three reform proposals are the immediate context for COLA. The *uniform lottery* (equal odds for all non-playoff teams) is the only mechanism satisfying Munro-Banchio's NTC; it abandons differential reward and the league has not adopted it. The *wheel* (a pre-committed rotation of draft positions across teams, attributed informally to Mike Zarren) eliminates tanking by decoupling record from draft odds, at the cost of removing the rebuilding mechanism for weak teams. The 2026 *3-2-1 lottery* [8] expands the lottery to 16 teams across four tiers (2:3:2:1 balls per tier from the worst-record tier down) and includes the play-in losers; structurally it is a uniform lottery with adjustments for the play-in and a reduction for the bottom three teams, accepting the Munro-Banchio tradeoff. COLA takes a different route: instead of flattening single-season odds, it shifts the priority basis from single-season records to multi-year playoff history, an axis neither Munro-Banchio nor 3-2-1 considers.

3 The COLA Mechanism Family: A Configuration Space

COLA is a family of mechanisms parameterized by seven design dials. Each NBA lottery reform proposal under recent discussion (the status quo weighted lottery, the 2026 "3-2-1" proposal, and each COLA variant proposed in Highley et al. [2] and the Substack series) corresponds to a specific choice of values for those seven dials. The framework is the league's design lever: a policymaker who wants to move a downstream property (anti-tanking strength, weak-team rebuild horizon, broadcast complexity, multi-year predictability) is moving one or more dials. We organize the section in four parts: the lottery index as the family's unified state variable (Section 3.1), the seven dials that complete the configuration (Section 3.2), the variant map populating the dials for each proposal under consideration (Section 3.3), and the mapping from dials to league-relevant properties (Section 3.4). Incentive compatibility for the family is closed by five assumptions [2], summarized in Section 3.5.

3.1 Lottery Index

Every COLA variant maintains a per-team, integer-valued state across seasons and uses that state to determine draft priority (DP) in the current season. We refer to this state as the *lottery index* and write $L_i^{(t)}$ for team i at the end of season t. Depending on the variant, $L_i^{(t)}$ may track drought length (consecutive seasons without a playoff series win or top-3 pick) or a stockpile of lottery tickets; both are manifestations of the same index.

Definition 1 (Lottery Index)**.** *For each team i and season t, the lottery index $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$ is determined by an initial value $L_i^{(0)}$, a per-season* increment $\Delta_i^{(t)}$ *(variant-specific, depending on team performance and eligibility), and a per-season* diminishment $\rho_i^{(t)} \in [0, 1]$ *that scales the index down based on playoff and draft outcomes:*

$$L_i^{(t+1)} = \left(L_i^{(t)} + \Delta_i^{(t)}\right) \cdot \left(1 - \rho_i^{(t)}\right). \tag{1}$$

The pool *in season t is $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$, where $E^{(t)} \subseteq \{1, \ldots, 30\}$ is the set of teams eligible for the lottery in that season.*

The three components of the update (the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$) are three of the seven dials that distinguish the variants. For a non-playoff team, the increment is positive (the team accumulates priority); for a team that wins a playoff series or a top draft pick, the diminishment is positive (the team consumes some or all of its accumulated priority). Section 3.2 formalizes these three components together with four additional dials that complete the parameterization; Section 3.3 maps each variant onto its 7-tuple in the resulting configuration space.

3.2 Seven Dials

Definition 1 identifies three dials that determine a variant's dynamics: the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$. Four additional dials, implicit in the prose definitions of the original paper [2] and the Substack series but not yet formalized as design degrees of freedom, complete the configuration: how draft positions are assigned given the index vector, whether the index is bounded above, how long the index persists across seasons, and how identical indices are resolved. We promote these to first-class components of a unified configuration.

Definition 2 (COLA Configuration)**.** *A* COLA configuration *is a 7-tuple $(E, \Delta, \rho, W, C, S, T)$ specifying:*

1. ***Eligibility*** $E^{(t)} \subseteq \{1, \ldots, 30\}$*: a season-$t$ predicate selecting the teams that participate in the lottery (per Definition 1).*

2. ***Increment*** $\Delta_i^{(t)} \in \mathbb{Z}_{\geq 0}$*: a non-negative integer added to the index of each eligible team, determined by season-t state (per Definition 1).*

3. ***Diminishment*** $\rho_i^{(t)} \in [0, 1]$*: the fraction of the index removed after team i's playoff or draft outcome (per Definition 1).*

4. ***Lottery weighting*** W*: a rule mapping the index vector* $L^{(t)}|_{E^{(t)}}$ *to a distribution over assignments of teams to draft positions* $\{1, 2, \ldots, |E^{(t)}|\}$*. The rule may be deterministic (e.g., assign by index rank) or random (e.g., draw pick 1 with probability* L_i/P*, then draw pick 2 from the remainder).*

5. ***Cap*** $C \in \mathbb{Z}_{>0} \cup \{\infty\}$*: an upper bound on* $L_i^{(t)}$*, enforced as* $L_i^{(t)} \leftarrow \min(L_i^{(t)}, C)$ *after each increment.* $C = \infty$ *denotes no explicit cap.*

6. ***Carry-over scope*** S*: the temporal persistence rule, one of* single-season *(recompute* $L_i^{(t)}$ *from season-*t *state alone, no cross-season memory),* unbounded multi-year *(accumulate without limit),* bounded multi-year *(accumulate up to* C*), or* reset-on-event *(clear the index when a specified event occurs).*

7. ***Tiebreak*** T*: a total ordering on* $E^{(t)}$ *that resolves cases where* $L_i^{(t)} = L_j^{(t)}$ *for two or more eligible teams.*

A COLA variant *is a specification of all seven dials.*

Definition 1 expresses the dynamics of $L^{(t)}$ in terms of dials 1–3; dials 4–7 govern how those dynamics translate into draft positions (W), constrain the index's range (C) and temporal scope (S), and resolve ties (T). The four additional dials are not novel mechanisms: each variant in Highley et al. [2] and the Substack series specifies a choice for each of them. Formalizing the four implicit choices as typed dials enables two operations: cross-variant comparison via a single configuration object, and dial-by-dial analysis of how each value affects each downstream property. Section 3.3 populates the configuration for each variant. Section 3.4 maps the dials to the league-relevant properties that drive the adoption decision.

3.3 Variant Map

Table 1 populates each of the seven dials for the six variants under consideration: the five COLA variants and the NBA's 2026 "3-2-1" proposal, included as a degenerate configuration (Section 3.6). Each row is a dial; each column is a variant. Cell values follow the verbatim specifications in Highley et al. [2], the Substack series [9, 10, 3], and Highley's reaction post on the 3-2-1 proposal [8].

The variants trade off explainability, anti-tanking strength, and per-series tanking exposure differently. Simple COLA [10] eliminates the lottery in favor of a deterministic ordering on drought, with eligibility extended to first-round losers; Simple Lottery COLA [10] layers the pre-2019 NBA weighted odds on top of Simple's drought ordering. Classic COLA is the original variant of Highley et al. [2] and the variant our manipulation bound (Section 4) directly analyzes. Countdown COLA [10] uses the McCarty number (drought $\times$ wins) and a nested-pool Monte Carlo draw that guarantees no team falls more than four spots below its priority rank. Capped COLA [3], introduced in response to the NBA's reported reluctance to adopt a flexible lottery line, adds a wins-based increment, a stockpile cap (MAX $= 150$), and a six-step playoff diminishment schedule; the cap bounds the per-series tanking incentive (Lemma 3). The 2026 NBA "3-2-1" proposal [8] sits at a degenerate corner: dials Δ, ρ, and C are vacuous, S collapses to single-season, and only E (a tiered 16-team pool) and W (a tiered ball allocation) carry the proposal's design. The collapse is the proposal's defining structural feature, not an incidental implementation choice; Section 3.6 elaborates.

3.4 Dial-to-Property Mapping

The configuration space of Definition 2 reduces a variant comparison to a tuple comparison. The comparison is informative only when each dial corresponds to a downstream property of the resulting mechanism. Four properties dominate the league-side adoption discussion: anti-tanking strength (how much within-season manipulation is foreclosed), weak-team rebuild help (how reliably the worst teams accumulate priority),

Dial	Simple	Simple Lottery	Classic	Countdown	Capped	3-2-1
E	22 (non-playoff + R1 losers)	22 (same)	14 (non-playoff)	22 (same)	exclusion pool	tiered 16
Δ	drought $+1$	drought $+1$	$\alpha = 1{,}000$	McCarty $d \cdot w$	wins w (play-in & below)	none
ρ	reset on series win	reset on series win	5-step, two channels	reset on advancement	6-step + 5-step, two channels	none
W	deterministic by drought	pre-2019 odds top-14, deterministic rest	L_i/P for picks 1–4, rank for 5–14	nested-pool MC, tickets $(6, 5, 4, 3, 2)$	L_i/P for picks 1–5, rank for 6+	tiered balls $(2/3/2/1)$
C	∞	∞	∞	∞	$\textsc{Max} = 150$	n/a
S	reset-on-event	reset-on-event	unbounded multi-year	reset-on-event	bounded multi-year	single-season
T	most wins	subsumed by W	subsumed by W	subsumed by W	most wins (at cap)	subsumed by W

Table 1: The seven dials, evaluated for each variant. Dial labels: E eligibility, Δ increment, ρ diminishment, W lottery weighting, C cap, S carry-over scope, T tiebreak (per Definition 2). The 3-2-1 proposal sits at a degenerate corner: Δ, ρ, C are vacuous and S is single-season, leaving only E and W as active dials.

complexity (how legible the mechanism is to fans and media), and predictability (how forecastable a multi-year team trajectory is for general managers and owners). Table 2 summarizes the mapping.

The mapping logic is direct. Three dials (E, Δ, C) bound the manipulator's pool share $p_{\max}$ and the size of any pool perturbation: a larger eligible set yields a larger total pool P, the increment Δ bounds the per-team per-season accumulation, and the cap C bounds the index outright. Two dials (ρ, S) govern multi-year accumulation: the diminishment schedule determines how quickly an index decays after success, and the carry-over scope determines whether accumulation persists at all (the dial whose non-single-season value escapes the Banchio and Munro [1] impossibility). The remaining two dials (W, T) translate the index dynamics into draft positions: W determines whether the high-index team gets pick 1 deterministically or in expectation, and T resolves cases where the index is silent. Anti-tanking strength and weak-team rebuild help load on E, Δ, ρ, C, S; complexity loads on E, W (the fan- and media-visible dials) and on S (the league-counsel-visible dial); predictability loads on ρ, S (the multi-year forecasting dials). A league that finds Classic too unbounded can move C to a finite value (the Capped move). A league that finds multi-year scopes administratively heavy can move S to single-season (the 3-2-1 move). The framework's value to a policymaker is that the choice among variants becomes a choice among dial settings, not among monoliths.

3.5 Incentive Compatibility

Highley et al. [2] establish incentive compatibility (IC) for Classic COLA under five assumptions:

1. **Playoff primacy:** the value to a team of advancing in the playoffs exceeds the value of any plausible draft-pick gain.

2. **Playoff advancement:** within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.

3. **Picks 5–14 negligible at the margin:** no team profits from manipulating its position among those slots through within-season effort.

Dial	Anti-tanking strength	Weak-team rebuild help	Complexity	Predictability
E	Sets the pool P in Theorem 2; narrower pools admit larger gain	Determines which weak teams accumulate priority	High if simple count, low if exclusion-rule	Stable if record-based; less so if exclusion-rule
Δ	Bounds the per-season marginal-decision gain	Flat α rewards uniformly; w-based rewards within-season effort	High if flat, lower if formulaic	High if record-based
ρ	Cliff size at each round transition; controls repeat-tanking returns	Restores priority to teams missing the playoffs again after success	Medium	Medium
W	Spread across rank positions; decouples position from index in part	Tilts allocation toward worst-record teams	High if deterministic, low if Monte Carlo	High if deterministic, lower if random
C	Bounds $p_{\max}$ and the per-series cost (Lemma 3)	Low cap compresses the priority signal	Medium	High once known
S	Escapes the Banchio and Munro [1] impossibility via multi-year scope	Multi-year scope rewards persistent weakness	Medium	Lower under multi-year scopes (legal/contracting)
T	Edge case for W-deterministic variants	n/a	High if simple	High

Table 2: Dial-to-property mapping. Cell entries describe the form the dial's influence takes; the dials and the variants of Table 1 together specify how to move from one league-relevant property to another by altering one or more dial values.

4. **Pool manipulation negligible:** no team can meaningfully shift its expected draft position by manipulating which other teams are in or out of the lottery pool.

5. **No traded-pick protections:** pick trades do not interact with diminishment in a way that creates manipulation incentives.

Assumptions (1)–(3) and (5) hold by construction or follow from straightforward arguments. Assumption (4) is supported by Highley et al. [2] via a $1{,}000$-season simulation, observing a maximum probability gain of approximately 3%. Section 4 converts this empirical observation into a closed-form analytic bound for the Classic configuration.

3.6 The 3-2-1 Proposal as a Degenerate Configuration

The NBA's 2026 reform proposal ("3-2-1"), slated for an owners' vote at the end of May 2026, expands the lottery to sixteen teams across four tiers and assigns ball counts of two, three, two, and one per tier respectively. In the configuration of Definition 2: dials Δ, ρ, and C are vacuous; the carry-over scope S is single-season; the eligibility set E is the tiered 16-team pool; the lottery weighting W is the tiered ball allocation; and the tiebreak T is subsumed by the random draw. Three dials (E, W, S) carry the proposal's design. In Highley [8]'s framing, the mechanism is "essentially the uniform lottery but with an adjustment for the play-in and a punishment for the bottom 3 teams." By assigning the worst three teams fewer balls than the middle seven, the proposal does not satisfy the premise of the Munro-Banchio counterexample (that worse teams have strictly better odds), and sidesteps the impossibility at the cost of abandoning the rationale for differential odds in the first place. COLA takes the opposite route: instead of flattening single-season odds, it shifts the priority basis from single-season records to multi-year playoff history through dial S in modes other than single-season.

4 Pool Manipulation Bound

4.1 Setup

Let $P = \sum_{i=1}^{n} L_i$ denote the total lottery pool, where L_i is team i's lottery index and n is the number of lottery-eligible teams. A team's probability of winning the first pick is $p_i = L_i/P$.

The *pool manipulation* channel operates as follows. Team i deliberately loses to a high-index opponent h near the playoff boundary, causing h to make the playoffs. Team h (with index L_h) exits the lottery pool and is replaced by a lower-index team ℓ (with index $L_\ell < L_h$). The pool shrinks by $\Delta = L_h - L_\ell > 0$.

Proposition 1 (Manipulation Gain). *The probability gain for team i from a pool swap of size $\Delta = L_h - L_\ell$ is $G_i = L_i \cdot \Delta/[P \cdot (P - \Delta)]$.*

Proof. Before manipulation, $p_i = L_i/P$. After the swap, the new pool is $P' = P - \Delta$ and $p_i' = L_i/P'$. The gain is $G_i = p_i' - p_i = L_i\big(1/(P-\Delta) - 1/P\big) = L_i \cdot \Delta/[P(P-\Delta)]$. □

When $\Delta \ll P$ (typically $\Delta/P \approx 4\%$ in our backtest), the gain factors into two ratios: the manipulator's pool share (p_i) and the fractional pool change (Δ/P). Both have structural ceilings, which we exploit below.

Indexing convention. The bound below uses the expression $L_i = T_i \cdot \alpha$, where T_i is team i's consecutive non-playoff seasons. For Classic COLA, this is exact for any team starting from a stockpile reset (a championship or top-5 lottery pick) and is the steady-state attractor under the diminishment schedule. A fully general bound that handles ticket inheritance from earlier playoff series under the full diminishment dynamics is in the broader paper.

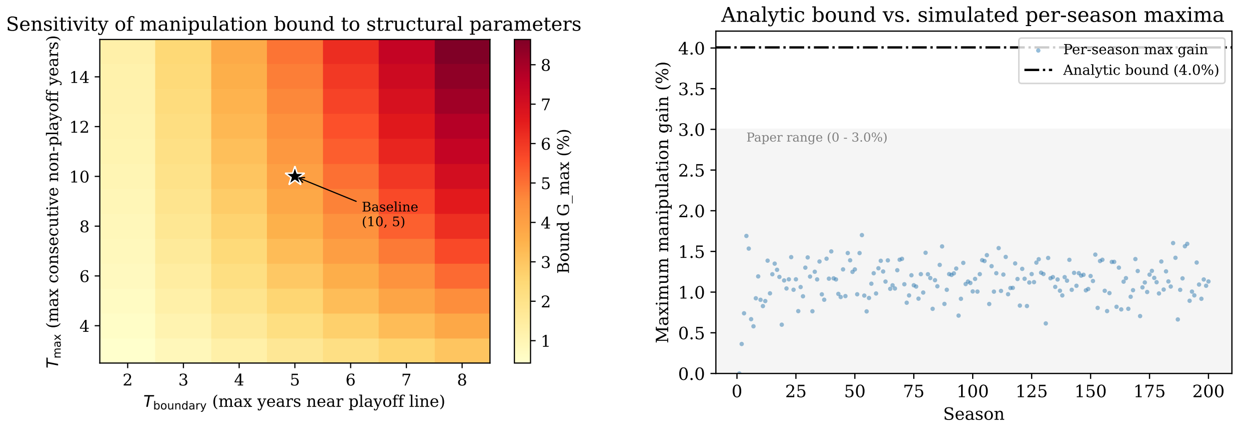
4.2 Conditioned Upper Bound

Definition 3 (Structural Anti-Correlation). *Let $T_{\max}$ denote the maximum consecutive non-playoff years for any team, and $T_{boundary}$ the maximum consecutive non-playoff years for a team near the playoff boundary (a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits* structural anti-correlation *when $T_{boundary} \ll T_{\max}$: teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.*

Theorem 2 (Conditioned Manipulation Bound). *Under structural anti-correlation with parameters $T_{\max}$, $T_{boundary}$, and average index multiplier $\bar{k} = \bar{L}/\alpha$, where α is the annual index increment, the maximum manipulation gain satisfies*

$$G_{\max} \leq \frac{T_{\max} \cdot (T_{boundary} - 1)}{(T_{\max} + n - 1) \cdot n \cdot \bar{k}}.$$

Proof sketch. The gain is maximized when L_i is maximized and Δ/P is maximized. *Bounding $p_{\max}$*: the highest-index team has $L_{\max} = T_{\max} \cdot \alpha$. The minimum pool size occurs when one team has $T_{\max}$ years of accumulation and the remaining $n-1$ teams each have the minimum α: $P_{\min} = (T_{\max} + n - 1) \cdot \alpha$. Thus $p_{\max} \leq T_{\max}/(T_{\max} + n - 1)$. *Bounding Δ/P*: the team being pushed out has at most $T_{\text{boundary}} \cdot \alpha$ tickets, and the team replacing it has at least α, so $\Delta_{\max} = (T_{\text{boundary}} - 1) \cdot \alpha$. In steady state, $P \approx n \cdot \bar{k} \cdot \alpha$. Combining via the first-order approximation yields the stated bound. □



(a) Bound vs. $T_{\max}, T_{\text{boundary}}$.

(b) Analytic bound vs. simulation.

Figure 1: Manipulation bound. (a) Sensitivity to the structural-anti-correlation parameters; the bound is most sensitive to T_{boundary}, and stays under 5% for $T_{\text{boundary}} \leq 5$. (b) Analytic bound (red) vs. the empirical distribution of manipulation gains across 200 Bradley-Terry-simulated seasons; the bound contains all simulated outcomes with margin.

Empirical calibration: We instantiate the bound with $T_{\max} = 10$ (the typical-case ceiling on consecutive non-playoff years across teams in our 26-season backtest), $T_{\text{boundary}} = 5$ (a conservative ceiling on the consecutive non-playoff years a near-boundary team can accumulate before improving enough to clear the boundary), $n = 14$ (the structural count of NBA non-playoff teams), and $\bar{k} = 3.1$ (an average team has roughly three years of carry-over before reset, matching the original paper's observation). These yield

$$G_{\max} \leq \frac{10 \cdot 4}{23 \cdot 14 \cdot 3.1} \approx 4.0\%,$$

which contains the simulation-observed maximum of 3.0% across $1{,}000$ seasons [2]. An independent 200-season Bradley-Terry simulation we ran confirms the bound, with observed maximum gain of 1.70%. The 26-season historical dataset contains one outlier: the Sacramento Kings' 16-year drought (2006–07 through 2021–22). Setting $T_{\max} = 16$ to include the outlier loosens the bound to approximately 5.1%, still well above the empirical extremes; we report the typical-case bound ($T_{\max} = 10$) as the headline because the next-longest droughts are Minnesota at 13 years and Phoenix and Charlotte at 10 each.

4.3 Sensitivity, Coalitions, and the Pick-Value Gap

The bound's parameter sensitivity (Figure 1a) shows that T_{boundary} is the dominant lever: even under pessimistic assumptions ($T_{\text{boundary}} = 7$), the bound stays under 7%, and under realistic conditions ($T_{\text{boundary}} \leq 5$) it stays under 5%. This matters operationally because T_{boundary} is the empirical quantity most amenable to direct measurement.

For completeness, coalitional manipulation (two or more teams coordinating to shrink the pool) does not amplify individual gains. Joint gain for a coalition $\{A, B\}$ is $(L_A + L_B) \cdot \Delta / [P(P - \Delta)]$. Even at 30% joint pool share, the gain remains $\leq 1.3\%$, and per-member gain cannot exceed the highest-index individual's gain. Coordination costs (detection risk, trust problems, sequential uncertainty) further dampen the channel.

The bound addresses the *probability* term of the manipulation incentive. The *value* term (the pick-value gap $d_k - d_{k+1}$ between adjacent draft positions) has no formal ceiling in the original specification, though it is bounded for any realistic draft class. Closing assumption (4) entirely would also require a separate argument capping $d_k - d_{k+1}$ via the distribution of draft-eligible prospect quality. We leave this to future work.

4.4 Per-Series Bound for Capped COLA

Capped COLA introduces a fourth manipulation channel: tanking through the playoffs themselves (a team within the cap declining to advance, in order to retain its stockpile). The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded above when stockpile grows large) to a fraction-of-MAX cost (bounded by construction).

Lemma 3 (Per-Series Tanking Cost)**.** *Let* MAX *be the Capped* COLA *stockpile cap. For any team entering the playoffs with* $L_i \leq \text{MAX}$*, the maximum reduction in* L_i *attributable to winning (rather than losing) any single playoff series is bounded by:*

$$\Delta L_{series}^{\max} = \begin{cases} 0.3 \cdot \text{MAX} & \textit{play-in advancers in round 1,} \\ 0.2 \cdot \text{MAX} & \textit{any other series advancement.} \end{cases}$$

At $\text{MAX} = 150$*, the bound is* 45 *tickets for the play-in case and* 30 *tickets for any subsequent series.*

Proof. Reading off Capped COLA's six-step diminishment schedule, the per-series increments in the diminishment fraction $\Delta\rho$ are: play-in advancer round 1 ($\rho_{\text{lose}} = 0.1$, $\rho_{\text{advance}} = 0.4$, $\Delta\rho = 0.3$); standard round 1 ($0.2 \to 0.4$, $\Delta\rho = 0.2$); round 2 to conference finals ($0.4 \to 0.6$); conference finals to NBA finals ($0.6 \to 0.8$); finals to championship ($0.8 \to 1.0$). All non-play-in transitions yield $\Delta\rho = 0.2$. The cost in tickets is $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \text{MAX}$, with equality at the cap. □

The bound makes the playoff-tanking incentive transparent at the league level: at $\text{MAX} = 150$, no single series win ever costs a team more than 45 lottery tickets, which is approximately a year and a half of accumulation for a team near the wins-increment ceiling. The cumulative regret for a finals-losing team at the cap is $0.8 \cdot \text{MAX} = 120$ tickets, decomposed into per-series increments per Lemma 3.

4.5 Three Closures of Assumption (4)

The pool-manipulation gap (assumption (4) in Section 3) admits at least three closure strategies, each represented in the literature.

- **Empirical closure:** Highley et al. [2] simulate 1,000 synthetic seasons under Bradley-Terry and observe a maximum gain of $\sim 3\%$. The closure is statistical: the assumption holds in the typical case but admits no parameter-explicit ceiling.
- **Behavioral closure:** the companion formal-proof video to [2] introduces an additional behavioral assumption $G >$ expected pool-manipulation gain (where G is the value of a single regular-season win), eliminating the channel via preference structure rather than mechanism design.
- **Analytic closure (this paper):** Theorem 2 provides a closed-form upper bound conditioned on structural properties of the index distribution. The bound is parameter-explicit and admits sensitivity analysis (Figure 1a).

The three closures compose: any practical COLA deployment can rely on whichever closure best matches the available evidence. Our contribution is the analytic option, which previously did not exist.

5 Empirical Validation

5.1 Simulation Model

The ZenGM Basketball-GM engine implements all five COLA variants and the standard NBA lottery; we use it as the canonical reference implementation. ZenGM's browser-coupled `createLeague()` path does not port cleanly to Node, so the testbed bypasses it and synthesizes regular-season game outcomes: strength-weighted win records and a single-elimination playoff bracket. The real ZenGM lottery code is unchanged: lottery draws use `draft.genOrder` and per-team `cola` updates from `cola.ts`.

Team strength persists across seasons via a Markov model tied to draft outcomes. Let $s_t \in [0, 1]$ denote a team's strength in season t, ρ the persistence parameter, μ the long-run parity mean, α the per-draft impact, σ the annual shock standard deviation, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ a Normal shock, and $\pi(p) = \max(0, (16 - p)/15)$ the value of a draft pick at position p (pick 1 contributes full α, pick 15 contributes zero, picks 16 and later contribute zero). The transition is

$$s_{t+1} = \mathrm{clip}\big(\rho\, s_t + (1 - \rho)\, \mu + \alpha\, \pi(p_t) + \varepsilon_t,\ 0,\ 1\big), \tag{2}$$

with $\rho = 0.9$, $\mu = 0.5$, $\alpha = 0.15$, and $\sigma = 0.05$. Initial strength is drawn from $\mathrm{Uniform}[0.3, 0.7]$. This preserves the feedback loop the primary objective (worst-franchise gap between conference-finals appearances) is designed to measure: a team's lottery position drives its draft pick, which drives its next-season strength.

The model is parametrically calibrated for plausibility, not fit to historical NBA team-strength trajectories. Sensitivity analysis on (ρ, α, σ), and a formal calibration against league rating systems or Basketball Reference strength-of-schedule data, is future work.

5.2 Backtesting Methodology

We compute COLA state for every team across 26 NBA seasons (1999–2025) using verified Basketball Reference data: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft-probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilize by approximately season 5; we report aggregates over 21 stable seasons (2005–2025).

Two methodological points warrant attention. First, an audit of all 364 lottery picks (14 picks $\times$ 26 seasons) identified 8 instances where the draft-day holder differed from the lottery-day holder via post-lottery trades. COLA requires *lottery-day* attribution: the team that held the pick when the lottery ran absorbs the diminishment, regardless of who selects. Second, the backtest is *static*: it applies COLA rules to actual historical outcomes. Under a real COLA regime, draft outcomes would differ in ways that change future drought lengths and indices. We caveat the comparative cross-variant claims accordingly.

5.3 Case Study: The Philadelphia "Process"

The 2013–14 through 2016–17 Philadelphia 76ers are the canonical strategic-tanking sequence in modern NBA history: four consecutive top-3 lottery outcomes (picks #3, #3, #1, #3) acquired through deliberate multi-season non-competitiveness. Under the current weighted lottery, each tanked season yielded a top-tier pick. Under Classic COLA, graduated diminishment makes the same sequence self-defeating. Table 3 shows Philadelphia's Classic COLA trajectory.

Two properties emerge. First, diminishing returns to repeat tanking: the Classic COLA diminishment schedule ($\{1.0, 0.75, 0.5, 0.25\}$ for picks #1–4) dominates the $\alpha = 1{,}000$ annual accumulation, so a team that receives a top-3 pick in year t cannot re-enter the top of the lottery in year $t + 1$ through tanking alone.

Season	W–L	Pick	COLA index	Rank
2013–14	19–63	#3	2,750	8/14
2014–15	18–64	#3	2,375	9/14
2015–16	10–72	#1	2,188	10/14
2016–17	28–54	#3	1,000	14/14

Table 3: Philadelphia 76ers under Classic COLA during the "Process." Each top-3 pick triggers graduated diminishment (#1: full reset; #3: 50% reduction), leaving Philadelphia at the bottom of the lottery pool by the fourth tanking season despite a 10–72 record. The Sacramento Kings hold the top Classic COLA index throughout this period, climbing from 7,250 in 2013–14 to 10,250 in 2016–17 via organic non-competitiveness.

MAX	Teams at cap (mean)	Teams at cap (max)	Sep. R1–R5 (mean)	Per-series cost (max)	Cum. regret (max)
75	6.71	9	0.9	22.5	45.0
100	4.33	8	11.7	30.0	53.7
125	2.86	4	27.9	37.5	53.7
150	**1.62**	**3**	**47.7**	**45.0**	**57.6**
175	0.95	2	65.4	52.5	67.2
200	0.52	2	76.4	60.0	76.8

Table 4: Capped COLA aggregates across 21 stable seasons (2005–2025). Per-series cost max = $0.3 \cdot \text{MAX}$ (Lemma 3); cumulative regret max is the worst empirical observation. The bolded MAX $= 150$ row is the variant's published default [3].

Second, pick attribution matters: the comparison is sensitive to who holds the pick on lottery day. The 2017 Boston-Philadelphia Fultz/Tatum swap was a post-lottery trade; under our convention, Boston's index absorbs the #1 diminishment, not Philadelphia's.

5.4 Capped COLA Cap Calibration

Capped COLA is parameterized by a single tunable constant MAX. The cap controls the trade-off between (i) ordering resolution at the top of the lottery (a higher cap permits more separation between the highest-stockpile teams) and (ii) per-series tanking exposure (Lemma 3: a higher cap raises the cost of advancing past round 1). Table 4 reports 21-stable-season aggregates from a sweep over MAX $\in \{75, 100, 125, 150, 175, 200\}$.

Three patterns emerge. Sub-100 caps over-compress the lottery: at MAX $= 75$, more than six teams sit at the cap on average and the rank-1 vs. rank-5 separation collapses to under one ticket, destroying the priority signal the family inherits from Classic. Caps above 175 erase the cap's purpose: fewer than one team sits at the cap on average, and the per-series exposure approaches the unbounded behavior of Classic. The published MAX $= 150$ choice produces a mean of 1.62 teams at the cap across the 21 stable seasons (range 0–3), matching the stated headline of "no more than three teams at the maximum at a time, and on average fewer than two" [3].

5.5 A Note on Trade Handling

A practical question for any deployment is how diminishment interacts with traded picks. Highley [11] identifies four options: (1) traded picks affect neither team's index; (2) the original holder's index absorbs the diminishment; (3) the receiving team's index absorbs the diminishment; (4) the diminishment is split. Each option has different incentive properties. Option (2) preserves the link between record and diminishment but over-penalizes teams that traded picks years before knowing they would be top-tier (e.g., the 2011 Clippers, who traded a future first that eventually became Cleveland's Kyrie Irving pick). Option (3) rewards the wrong actor when the recipient did not earn the pick through their own record. Highley [11] recommends option (2) paired with a strengthened Stepien Rule (a cap on the number of future firsts a team can trade), which limits the frequency at which option (2)'s over-penalization surfaces. We do not extend the analysis here; we flag it as a practical implementation question that any COLA deployment will face.

6 Discussion and Open Directions

The bounds in this paper formalize the manipulation-resistance properties of the COLA family. Theorem 2 bounds the open assumption (4) in Classic COLA's incentive-compatibility argument at approximately 4% under empirically calibrated parameters, with explicit sensitivity to $T_{\max}$, T_{boundary}, n, and $\bar{k}$. Lemma 3 bounds the per-series exposure that Capped COLA's stockpile cap was designed to address. Combined, the four manipulation channels available under Capped COLA (single-season tanking, pool manipulation, multi-year strategic tanking, and per-series playoff tanking) compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

6.1 3-2-1 as 2026 Stopgap, COLA as 2029 Candidate

The NBA's 2026 "3-2-1" proposal [8] is structurally a uniform lottery with an adjustment for the play-in tournament and a reduction for the three teams at the bottom of the standings. The proposal includes a three-year re-evaluation provision; the design discussion is expected to reopen in 2029. We read 3-2-1 as a short-horizon stopgap that accepts the Munro-Banchio tradeoff (uniform-style flattening, with the worst teams given fewer balls than the middle of the lottery pool). COLA is the natural candidate for the 2029 re-evaluation on a different design axis: dial S (carry-over scope) shifts the priority basis from single-season records to multi-year playoff history, an option neither 3-2-1 nor any single-season mechanism considers.

The choice between 3-2-1 and COLA is the choice between two answers to the Munro-Banchio impossibility. 3-2-1 sacrifices differential reward for non-tanking. COLA preserves differential reward across years by relocating it to a non-manipulable basis. The 2029 discussion will benefit from having both options analytically characterized.

6.2 Dial-Space Adoption Tradeoffs

Each dial of Section 3 matters more to some audiences than others. Three audience groups frame the adoption tradeoffs differently.

Fans and media care most about E (eligibility) and W (lottery weighting). The current 14-team weighted lottery fits a 30-second broadcast graphic; Capped's exclusion-rule pool and Countdown's nested Monte Carlo do not. Variants that simplify the eligibility rule (Simple, Classic) are easiest at this layer.

General managers and owners care most about Δ (increment) and ρ (diminishment). A multi-year rebuild requires a forecastable trajectory: Classic's flat $\Delta = \alpha$ and graduated ρ are the most tractable. Capped's wins-based Δ and six-step ρ add complexity but remain forecastable. Countdown's $\Delta = d \cdot w$ multiplies two factors that move in opposite directions for a rebuilding team and produces noisier multi-year forecasts.

League counsel cares most about S (carry-over scope). Multi-year scopes introduce questions that single-season scopes (3-2-1, the legacy lottery) avoid: how does mid-season team relocation, ownership change, or league realignment interact with carry-over priority? Reporting requirements and ambiguity both grow with S. The NBA's apparent preference for single-season-scope mechanisms in 3-2-1 is consistent with S being the dial under the most external pressure.

Open directions: three follow-on questions extend this work.

- *Tighter manipulation bound:* the structural anti-correlation in Definition 3 is a calibrated assumption rather than a formally derived consequence of the mechanism. A stationarity argument that bounds T_{boundary} as a function of league parameters (number of teams, season length, playoff structure) would close assumption (4) entirely.
- *Pick-value gap:* combining the probability bound (this paper) with a separate bound on the pick-value gap $d_k - d_{k+1}$, derived from the distribution of draft-eligible prospect quality, would close the manipulation incentive in absolute terms, not just in probability.
- *Axiomatic embedding:* the lift from Coreno and Balbuzanov [4]'s axioms (RP, EF1, SP) to multi-period priority mechanisms remains open. Section 2.2 flags the question; a formal verification would embed COLA in the assignment-mechanism literature and clarify which one-period impossibilities transfer to multi-period settings.

A reproducibility artifact accompanies this paper: an interactive backtester implementing all five COLA variants plus the 3-2-1 proposal across 26 NBA seasons, with locked-seed Monte Carlo for Countdown COLA probabilities [12].

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